

Merrill Dmonte

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🌐 [Portfolio](#) | [merrill-dmonte](#) | [Merrill04](#)

PROJECTS

E-Waste Recycling Management System [Next.js](#) · [React.js](#) · [MongoDB](#) · [Scikit-learn](#) [GitHub](#) Apr 2025

- Developed a full-stack e-waste recycling platform using **Next.js App Router**, MongoDB, JWT Authentication, and REST APIs, enabling secure user registration, booking management, and facility discovery.
- Integrated **Mapbox GL JS**, Geolocation API, Mapbox Directions API to provide nearest recycling facility search, interactive maps, and optimized navigation routes.
- Built an **AI-powered price prediction system** using Python, Scikit-learn Random Forest, Next.js API Routes, and Google Gemini API to estimate recycling value and provide an intelligent e-waste assistance chatbot.

URL Shortening Service [Node.js](#) · [Express.js](#) · [PostgreSQL](#) · [Redis](#) · [React.js](#) · [JWT](#) [GitHub](#) July 2026

- Built a full-stack URL shortener with **Node.js/Express REST API** and React frontend, implementing JWT authentication with access/refresh token rotation and bcrypt password hashing for secure multi-user session management.
- Engineered a **Redis caching layer** on the redirect endpoint reducing database load, with PostgreSQL as persistent store using indexed short-code lookups, db-migrations runner, and cascaded relational schema across users, URLs, and click events.
- Developed a **click analytics pipeline** capturing device type, browser, and referrer per redirect via async UA parsing, aggregated using PostgreSQL window functions and served as daily/weekly/monthly timeseries with Recharts visualizations on the frontend.

EDUCATION

Fr. Conceicao Rodrigues College of Engineering, Mumbai University 2022 – 2026

BE Computer Engineering — CGPA: 8.01

- **Coursework:** Data Structures, Computer Networks, Operating Systems, DBMS, Machine Learning

SKILLS

Languages : JavaScript, TypeScript, Java, C++, Python

Backend : Node.js, Express.js, REST APIs, JWT Authentication

Frontend : React.js, Next.js, Tailwind CSS

Databases : PostgreSQL, MongoDB, Redis

Cloud & DevOps : AWS (EC2, S3, Lambda), Docker, Git, Postman

Concepts : Data Structures & Algorithms, System Design, OOP, Microservices, CI/CD

CERTIFICATIONS & ACHIEVEMENTS

AWS Academy Graduate — [Cloud Architecting - Training Badge](#)

AWS Academy Graduate — [Machine Learning for Natural Language Processing - Training Badge](#)

AWS Academy Graduate — [Microservices and CI/CD Pipeline Builder - Training Badge](#)

AWS Academy Graduate — [Cloud Web Application Builder - Training Badge](#)

Solved 200+ DSA problems on [LeetCode](#) and [Codeforces](#)